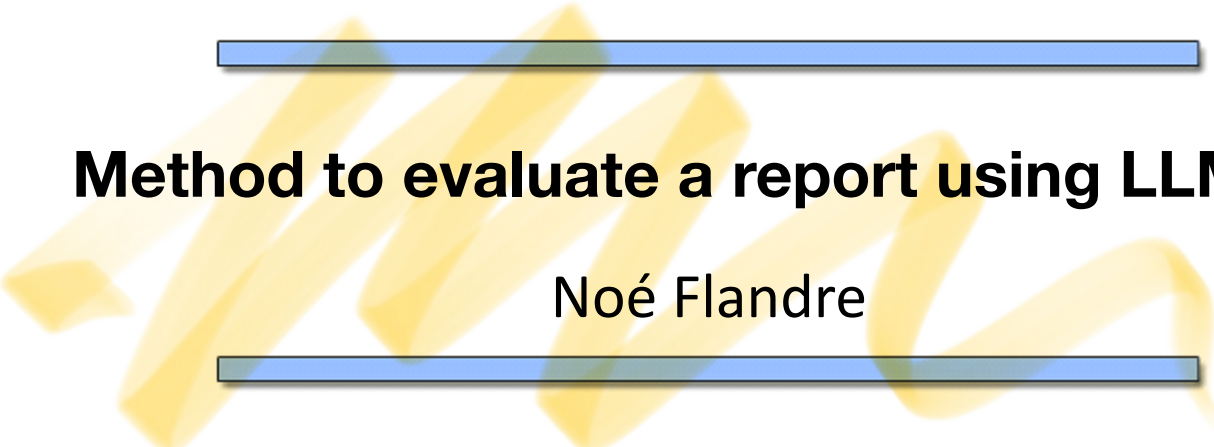
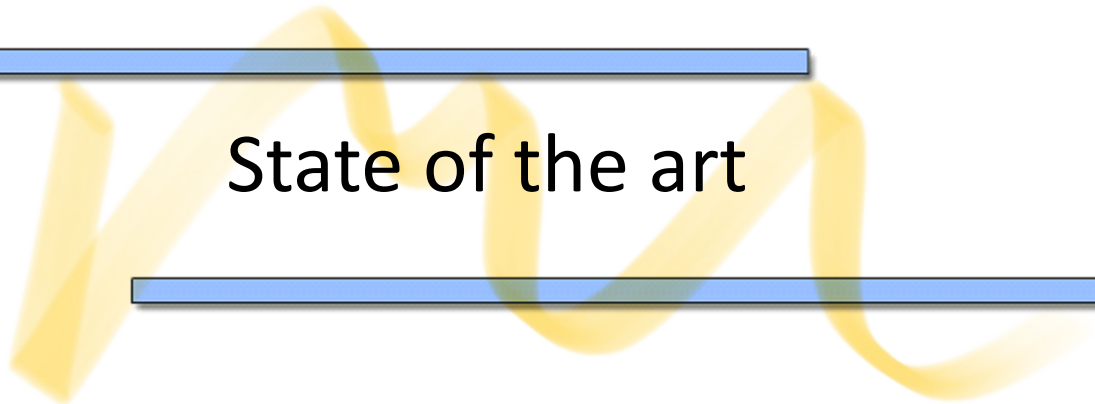




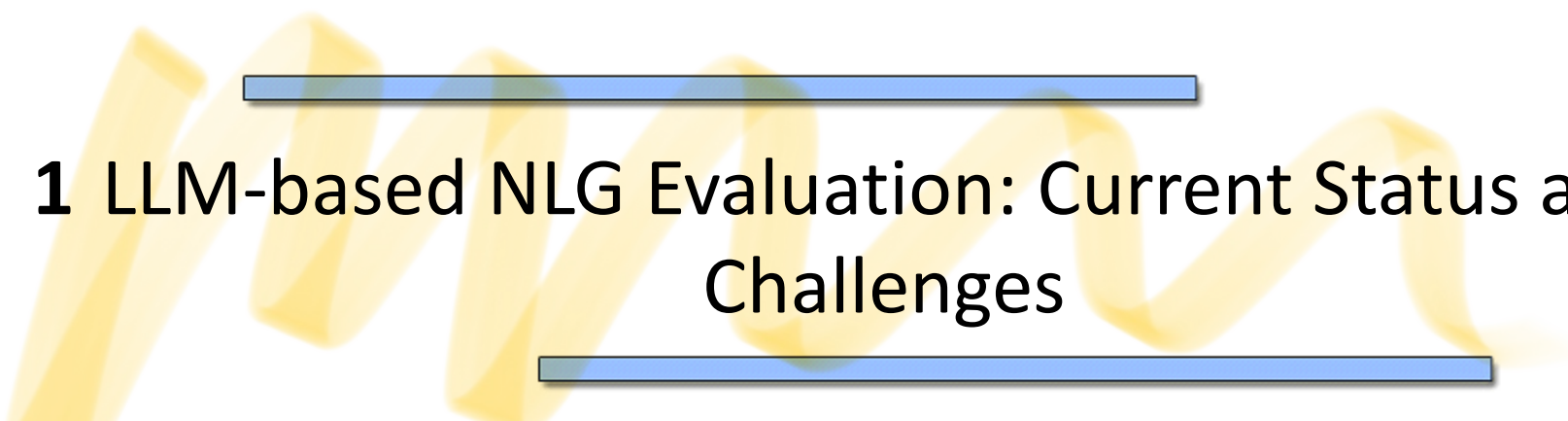
Method to evaluate a report using LLMs

Noé Flandre

1



State of the art



1 LLM-based NLG Evaluation: Current Status and Challenges

LLM-derived Metrics

Embedding-based Metrics

Example : BERTScore

Need a reference

LLM-derived Metrics

Probability-based Metrics

Example : BARTScore

Look at how likely a model is to generate the given text.

There are application in Hallucination Detection

LLM-derived Metrics

Pros & Cons

Strong correlation with human judgement

Lack of robustness

Time consuming and require more computational resources

More biased than traditional metrics

Prompting LLMs

Evaluation Methods

Scoring using a Likert scale from 1 to 5

Pairwise comparison (better for medium sized LLMs)

List-wise ranking

Boolean Q&A

Error Analysis based on predefined error categories

Prompting LLMs

Task Instructions

Form and requirements : different templates & lengths

Asking for analysis and explanations : higher correlation with human judgment

In context examples : simple one shot doesn't improve the performance while iterative updated example does

Prompting LLMs

Input content

For certain evaluations like faithfulness, the source document is needed as well as the target

Some references can be provided

Some context and relevant facts can be provided

Prompting LLMs

Evaluation criteria

Purely human criterias

Sees of human criterias and let the LLM create its own

Using subcategories

Using chain of aspects

Prompting LLMs

Role & Interaction

Chain style interaction : analysis, ranking & scoring

First stage of event units or questions from the target, then answering the questions and checking if they meet the requirements

Prompting LLMs

Role & Interaction

Network-style interaction : debates among multiple LLMs with different communication strategies

Decomposing the problems, solving them and aggregating the answers

Prompting LLMs

Role & Interaction

Different LLMs : Pair wise comparison and then discussion for the final result

One LLM who generated the text is the examinee and the other one the examiner

Prompting LLMs

Pros & Cons

Flexibility in the prompt

Correlation with human judgment

Biased : position bias, favor longer and verbose responses as well as responses from themselves

Prefer factual errors rather than grammatical errors

Prompting LLMs

Pros & Cons

Lower correlation with human assessments when scoring high quality summaries

Used by most of the existing work

Fine-tuning LLMs

Data construction

Difficulty in constructing large scale datasets

Fine-tuning LLMs

Evaluation Method

Highly diversified

Criteria being imposed or determined by the LLM itself

Fine-tuning LLMs

Fine Tuning Implementation

A lot of different processes

Fine-tuning LLMs

Pros & Cons

Low cost inference

Good reproductibility

Performance close to GPT4

Tackling some biases

Iterative improvement

Computational expenses

Base LLM often outdated as the open source models are rapidly being updated

Human-LLM Collaborative Evaluation

Scoring and Explaining

99% of accuracy with half human effort

LLMs generate initial ideation, humans engage in scrutiny

Scores along with their explanations for collaborative evaluation

Human-LLM Collaborative Evaluation

Pros & Cons

Efficiency and Cost Balance

Complementary Strengths

The most promising method according to the authors

Sensitivity to Query Formats

Confidence Level Assessment: Current methods struggle to accurately assess when to trust the LLM.

2 A Field Guide to Automatic Evaluation of LLM-Generated Summaries

Goals

Evaluating the summary on different qualities :
Fluency, coherence, relevance, factual consistency,
fairness, similarity in content and style

Methods

Reference-based Metrics

N-gram based metrics : BLEU, ROUGE, JS2

Embedding Based Metrics : BERTScore, MoverScore, SMS

Limitations : Poor correlation with human evaluators, lack of interpretability, inherent bias, inability to capture subtle nuances in language

Methods

Reference-free Metrics

Quality-based Metrics : SUPERT, BLANC, ROUGE-C

Entailment-based Metrics : SummaC, FactCC, DAE

Factuality, QA and QG-based metrics : SRLScore, QAFactEval, QuestEval

Better correlation to human evaluators, but bias against their underlying models' outputs and against higher-quality text

Methods

LLM-Based Evaluators

Prompt-based evaluators : summary alone (reference-free), summary, original text, and potentially topic for summarization (reference-free), summary and ground-truth (reference-based)

e.g : RTS, MCQ, H2H, G-Eval

Limitations : positional bias, verbosity bias, self enhancement bias, limited mathematical and reasoning capabilities

Methods

LLM-Based Evaluators

LLM Embedding-base metrics : embedding-ada-002 are being used

Methods

Practice to avoid pitfalls

Developing a suite of metrics rather than one to rule them all

Use custom metrics for particular use case and validate them against human evaluated summaries : e.g semantic difference between 8GB VS 16GB

Combining LLM based and smaller PTMs based metrics / non-model metrics : allow to detect discrepancies

Methods

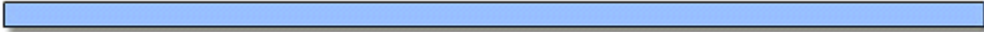
Practice to avoid pitfalls

Visualizing the metrics and analyzing them

Involve the end users experts and design metrics based on their own way to judge the quality of the summary



3 EvaluLLM: LLM Assisted Evaluation of Generative Outputs



State of the Art

LLMs fine tuned for evaluation

Prometheus : Pearson correlation of 0.897 with human evaluators

EvaluLLM

Task and Dataset Definition: Evaluable evaluates NLG tasks by defining a task prompt (P) and an input dataset (D). Variables from the dataset can be embedded in the task prompts using standard curly bracket notation.

Model Evaluation: The system runs the task prompts through various LLMs (evaluated LLM E) and uses a user-defined evaluation criterion (C) to assess the outputs. The criterion is a natural language description of quality specific to the user's needs.

Comparison and Scoring: Outputs are compared in pairs to reduce complexity, with a focus on relative preferences rather than absolute scores. Evaluations can be either local (per instance) or global (aggregated over multiple instances).

EvaluLLM

Interface Design: The Evaluable interface has two main components: a generator for embedding task prompts and a reviewer for selecting and evaluating outputs. Users can manually review and rate a sample of outputs or rely on automatic evaluations.

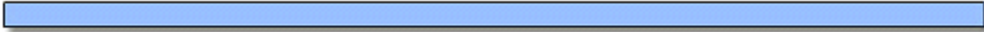
Agreement Score: The system calculates an agreement score to measure how well the user and the evaluator LLM agree on output quality, aiding in understanding and debugging evaluation criteria.

EvalLLM

Limitations : All the limitations from an LLM. They suggest to not replace humans with it but to make the evaluation less laborious.



4 Themis: Towards Flexible and Interpretable NLG Evaluation



Themis

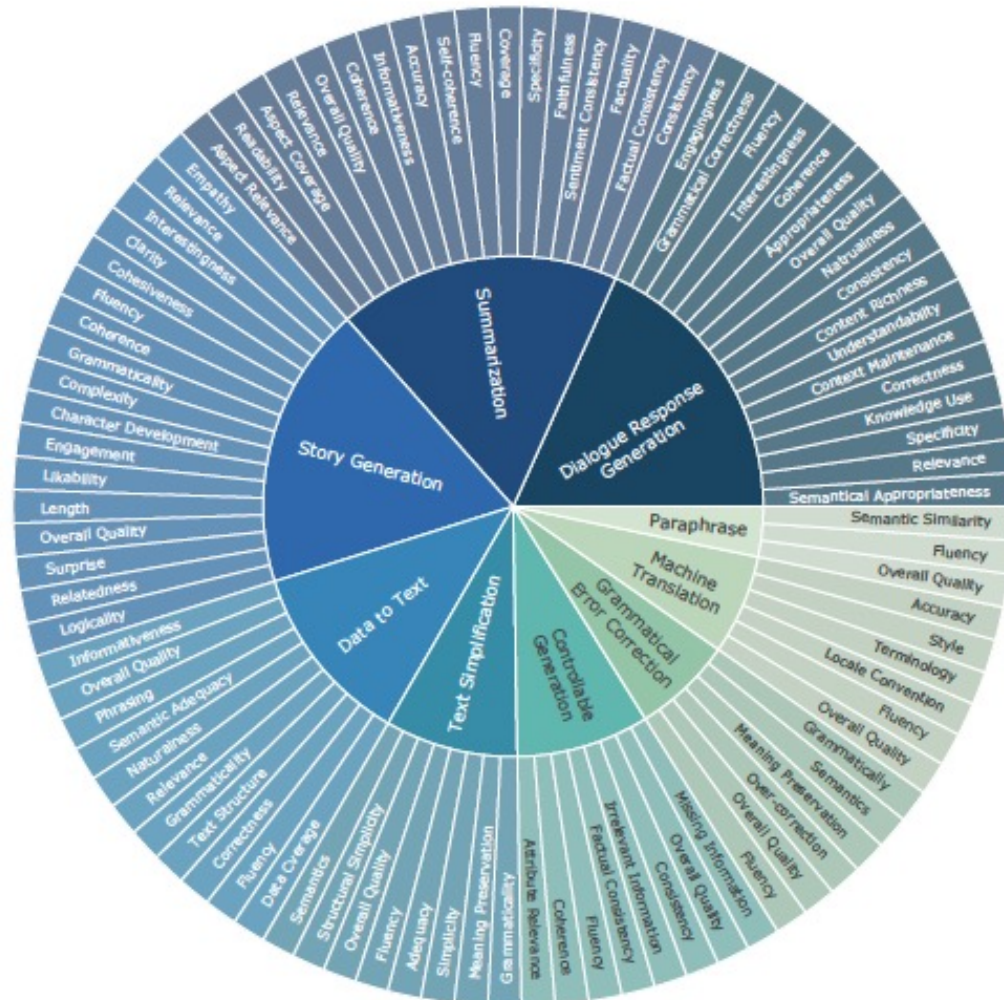
Broad application range

Reference-free

Interpretability thanks to the given explanations along with scores

Method	Vers.	Inde.	Flex.	Inte.	Open.
UniEval	✗	✗	✓	✗	✓
G-Eval	✓	✓	✓	✓	✗
X-Eval	✓	✗	✓	✗	✗
Prometheus	✓	✗	✓	✓	✓
Auto-J	✓	✓	✗	✓	✓
InstructScore	✓	✗	✗	✓	✓
TIGERScore	✓	✓	✗	✓	✓
Themis (Ours)	✓	✓	✓	✓	✓

Themis





Questions break

